

Internship Application Tracker - Project Proposal

Team 7: Sid, RaMar Wilson, Ihor

Course: Database Management

Professor: Dr. Forouraghi

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Saint Joseph's University

Project Description

The **Internship Application Tracker** is a comprehensive web-based application designed to help students efficiently manage their internship application process. Students often apply to dozens of companies and struggle to keep track of application statuses, interview schedules, recruiter contacts, and follow-up tasks. Our application addresses this challenge by providing a centralized, cloud-based platform that streamlines the entire application management workflow.

The system will enable students to track all aspects of their job search, from initial application submission through final offer acceptance. By automating data entry through Gmail integration and providing visual analytics, students can focus on preparation and follow-up rather than administrative tracking. The application will be accessible from any device with internet connectivity, ensuring students can update their information on the go.

Technical Stack

Frontend: Next.js 16 with React 18, Tailwind CSS

Backend: Next.js API Routes (Node.js runtime)

Database: MySQL 8.0 (Cloud-hosted on Railway)

Deployment: Vercel (with CI/CD pipeline)

Authentication: OAuth 2.0 for Gmail, session-based authentication

Architecture: 3-tier (Browser → Node.js → MySQL)

Database Design

The application will utilize a **7-table normalized database** with the following structure:

1. **Users** - Store user account information and credentials
2. **Applications** - Track all internship applications with company details
3. **Interviews** - Manage interview schedules and outcomes
4. **Contacts** - Store recruiter and company contact information
5. **Reminders** - Track follow-up tasks and deadlines
6. **Documents** - Manage resumes, cover letters, and supporting materials
7. **Notes** - Store additional observations and research about applications

All tables will maintain referential integrity through foreign key constraints with CASCADE operations.

Core Functionalities & SQL Operations

1. Application Management (CRUD Operations)

SELECT Operations:

- View all applications with sorting and filtering by status, company, or date
- Search applications by company name or position title
- Retrieve specific application details by ID
- Display recent applications on dashboard (last 5)

INSERT Operations:

- Add new internship applications manually
- Import applications automatically from Gmail
- Batch import multiple applications from parsed emails

UPDATE Operations:

- Modify application status (Applied → Interview → Offer → Accepted/Rejected)
- Edit application details (company, position, location, salary, notes)
- Update application information inline from table view

DELETE Operations:

- Remove applications with confirmation dialog
- Cascade delete related interviews, contacts, and notes automatically

2. Interview Scheduling & Tracking

SELECT with JOIN:

- Retrieve all interviews with associated application details
- Display upcoming interviews with company and position information
- View interview history with outcomes

INSERT:

- Schedule new interviews linked to specific applications
- Record interview type (Phone Screen, Technical, Behavioral, Panel, Final)
- Store interviewer contact information and location

UPDATE:

- Modify interview date, time, or location
- Update interview outcome (Pending, Passed, Failed, Cancelled)

DELETE:

- Cancel interviews with automatic cleanup

3. Analytics Dashboard

SELECT with Aggregations:

- Calculate total applications count: `SELECT COUNT(*) FROM Applications`
- Compute success rate: `(Offers / Total Applications) * 100`
- Determine interview conversion rate: `(Applications with Interviews / Total) * 100`
- Calculate average response time: `AVG(DATEDIFF(updated_at, application_date))`

SELECT with GROUP BY:

- Applications by status: `GROUP BY application_status`
- Top companies: `GROUP BY company_name ORDER BY COUNT(*) DESC`
- Monthly trends: `GROUP BY DATE_FORMAT(application_date, '%Y-%m')`

4. Contact Management

CRUD Operations for Recruiters:

- Store recruiter names, titles, emails, phone numbers, and LinkedIn profiles

- Link contacts to specific applications
- Track communication history

5. Reminder System

Task Management:

- Create follow-up reminders for applications
- Set interview preparation deadlines
- Schedule thank-you note reminders
- Mark reminders as completed

6. Gmail Integration

Automated Import:

- OAuth 2.0 authentication with Google
- Parse emails for application confirmations
- Extract company names, position titles, and dates
- Bulk import selected applications

7. User Authentication

Session Management:

- User registration with encrypted password storage
 - Secure login/logout functionality
 - Protected routes requiring authentication
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Additional Features

Search & Filter

- Real-time search across company names and position titles
- Filter applications by status (All, Applied, Interview, Offer, Accepted, Rejected)
- Sort by application date, company name, or status

Visual Analytics

- Statistics cards showing total applications, pending, interviews, and offers
- Progress bars for applications by status
- Bar charts for monthly application trends
- Ranked list of top companies by application volume

User Experience

- Responsive design for mobile, tablet, and desktop
 - Loading states with spinner animations
 - Success/error notifications with auto-dismiss
 - Inline editing for quick updates
 - Confirmation dialogs for destructive actions
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Security & Data Integrity

- **SQL Injection Prevention:** Parameterized queries throughout
 - **Password Security:** Bcrypt hashing for stored passwords
 - **Foreign Key Constraints:** Maintain referential integrity
 - **Input Validation:** Server-side and client-side validation
 - **OAuth 2.0:** Secure third-party authentication
 - **HTTPS:** Encrypted data transmission in production
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Expected Outcomes

Upon completion, students will be able to:

1. Track unlimited internship applications in one centralized location
 2. Schedule and manage interview rounds with automated reminders
 3. Visualize application success rates and identify improvement areas
 4. Import applications automatically from Gmail, saving time on data entry
 5. Access their data from any device with internet connectivity
 6. Generate insights about their job search patterns and effectiveness
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Project Timeline

Week 1-2: Database design, ER diagram, schema implementation

Week 3-4: Backend API development with all SQL operations

Week 5-6: Frontend development with React and Next.js

Week 7: Gmail OAuth integration and email parsing

Week 8: Testing, deployment, and documentation

Team Member Responsibilities

Ihor: Database architecture, ER diagram, schema design, SQL DDL statements

RaMar Wilson: Frontend development, API integration, Gmail OAuth, cloud deployment

Sid: Backend APIs, complex SQL queries, testing, quality assurance